

DoNOF

DoNOF is a computational chemistry code based on Natural Orbital Functional Theory (NOFT). The original code started on January 21, 2009 as PNOFID.

Current functionality includes:

- PNOF5, PNOF6, PNOF7, and GNOF
- Single-point energies, gradients, geometry optimization, numerical Hessians, transition-state optimization, and molecular dynamics
- ERPA, post-NOF corrections, and nonlinear optical properties

Build

Requirements:

- A Fortran compiler compatible with the current **Makefile**
- **libcint**
- OpenMPI, if MPI execution is needed

Library source:

- <https://github.com/sunqm/libcint>

Build targets:

```
make serialg    # exe/DoNOFg.x
make ompg       # exe/DoNOFompg.x
make mpig       # exe/DoNOFmpig.x
```

Run

Example inputs are available in **examples/**.

Available wrapper scripts:

```
./run_donofg filename
./run_donofompg filename
./run_donofmpig filename
./run_donofg_dyn filename
./run_donofompg_dyn filename
./run_donofmpig_dyn filename
```

Input is controlled mainly through **&INPRUN** and **&NOFINP**. Molecular-dynamics jobs also use **&INPDYN**.

Documentation

- Online documentation: <https://donof-documentation.readthedocs.io>
- Local namelist reference: **doc/namelist.md**
- Local PDF version: **doc/namelist.pdf**

Contact

DoNOFsw@gmail.com